C. Decreasing String

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Recall that string a is lexicographically smaller than string b if a is a prefix of b (and $a \neq b$), or there exists an index i ($1 \le i \le \min(|a|, |b|)$) such that $a_i < b_i$, and for any index j ($1 \le j < i$) $a_j = b_j$.

Consider a sequence of strings $s_1, s_2, \dots, s_n$, each consisting of lowercase Latin letters. String s_1 is given explicitly, and all other strings are generated according to the following rule: to obtain the string s_i, a character is removed from string s_{i-1} in such a way that string s_i is lexicographically minimal.

For example, if $s_1 = \mathrm{dacb}$, then string $s_2 = \mathrm{acb}$, string $s_3 = \mathrm{ab}$, string $s_4 = \mathrm{a}$.

After that, we obtain the string $S = s_1 + s_2 + \dots + s_n$ (S is the concatenation of all strings $s_1, s_2, \dots, s_n$).

You need to output the character in position pos of the string S (i. e. the character S_{pos}).

Input

The first line contains one integer t — the number of test cases ($1 \le t \le 10^4$).

Each test case consists of two lines. The first line contains the string s_1 ($1 \le |s_1| \le 10^6$), consisting of lowercase Latin letters. The second line contains the integer pos ($1 \le pos \le \frac{|s_1| \cdot (|s_1|+1)}{2}$). You may assume that n is equal to the length of the given string ($n = |s_1|$).

Additional constraint on the input: the sum of $|s_1|$ over all test cases does not exceed 10^6.

Output

For each test case, print the answer — the character that is at position pos in string S. **Note that the answers between different test cases are not separated by spaces or line breaks**.

Example

input

```
3
cab
6
abcd
9
x
1
```

output

```
abx
```